

AI/ML Intelligence Layer (Layer 4)

Technical Architecture & Algorithmic Design Specification

DOCUMENT ID SUBSENSE-TDD-ML-004	SYSTEM LAYER Layer 4: Analytics & ML Stack	TARGET ENVIRONMENT Hybrid Edge (ESP32) + Cloud (K8s)	COMPLIANCE MANDATE DGMS Mine Safety Circulars
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1. PURPOSE & SCOPE

The **AI/ML Intelligence Layer** constitutes the real-time analytical core of the **SubSense** Smart Mine Subsidence Monitoring Platform. Positioned at *Software Stack Layer 4*, it ingests high-frequency, multi-sensor time-series telemetry streamed from underground and surface wireless mesh nodes (via gateway aggregators) and converts raw, noisy measurements into actionable, explainable, and spatially-grounded subsidence risk intelligence.

In bord-and-pillar extraction, longwall coal mining, and metalliferous stoping environments, surface and subsurface deformation exhibits non-linear, multi-phase mechanics (initial bending, catastrophic bed separation, strata caving, and compaction). Purely statistical thresholding triggers excessive false alarms from heavy earthmoving machinery, blast vibration, and thermal shifts, leading to alarm fatigue. Conversely, delayed detection risks human lives, structural collapse of surface infrastructure, and catastrophic haul-road failures.

This specification establishes the exhaustive technical design of the algorithmic pipelines, data schemas, edge-cloud partitioning, and verification metrics across ten modular engines:

- **Unsupervised Per-Node Anomaly Engine:** IsoForest + Autoencoder
 - **Deformation Forecaster:** Seq2Seq LSTM with Uncertainty Quantiles
 - **Spatial Interpolation:** Universal Kriging with Subsurface Panel Drift
 - **Cross-Sensor / Cross-Node Correlation Engine:** Spatio-Temporal Filtering
 - **Graph Correlation Model:** Mesh-Topology Aware Graph Attention (GAT)
- **Explainability & Confidence Layer:** SHAP & Attention Attribution
 - **Predictive Time-to-Critical:** Trajectory Projection Countdown
 - **Feedback Loop:** Human-in-the-Loop Active Retraining
 - **TinyML Edge Deployment:** 8-bit Quantized On-Device Inference
 - **InSAR Macro-Fusion:** Sentinel-1 Coarse-Scale Baseline Cross-Check

2. CORE ARCHITECTURAL DESIGN PRINCIPLES

- #### 1. Defense in Depth over Single-Model Trust

No single algorithm is authorized to escalate an incident to warning status. Point anomalies, temporal progression, multi-sensor corroboration, and graph spatial diffusion must vote cooperatively. Triangulation across independent mathematical formulations virtually eliminates false trips from rock drills, haul trucks, or lightning.
- #### 2. Edge-First, Cloud-Enhanced Partitioning

Underground mine environments frequently experience mesh network partitioning and backhaul severance. Every critical detection routine features a zero-dependency TinyML footprint deployable directly to microcontrollers/gateways for autonomous local horn/siren actuation, while cloud engines run deep spatial synthesis.

3. Explainability by Construction

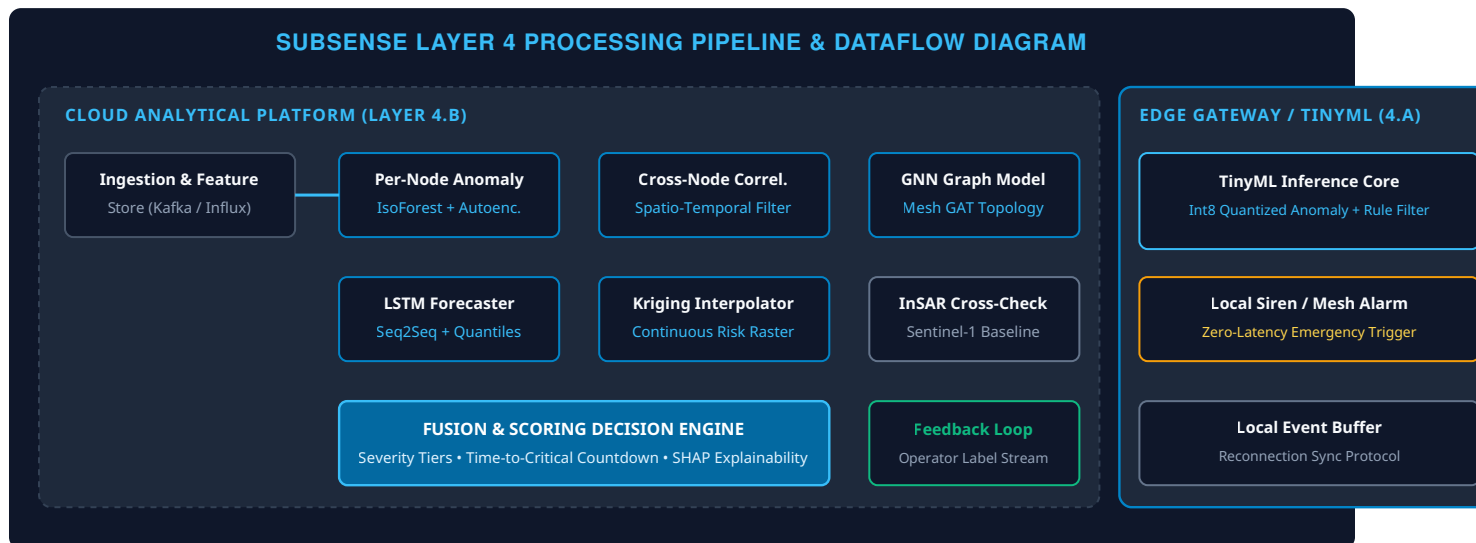
Black-box classifications are explicitly rejected by mining safety inspectorates (DGMS, MSHA). Every risk score generated by the layer emits normalized sensor feature attributions, contributing neighbor node IDs, and plain-language operational summaries alongside raw tensor outputs for evidentiary audibility.

4. Physically-Grounded Spatial Geometries

Interpolation and message-passing are strictly constrained by the physical mine plan. Rather than isotropic planar assumptions, edge weights and spatial drift equations encode goaf extraction lines, bord width, pillar dimensions, fault planes, and overburden stratigraphic thickness.

3. HIGH-LEVEL AI/ML SYSTEM ARCHITECTURE

The dual-tier architecture bridges edge sensor nodes and high-throughput cloud analytical infrastructure. The cloud layer handles distributed stream processing, deep feature synthesis, spatial rasterization, and automated retraining loops, while edge gateways sustain fail-safe autonomous monitoring.



4. FORMAL DATA CONTRACTS & SCHEMAS

Standardized payload schemas guarantee interoperability between the mesh transport layer (Layer 3), the analytics orchestrator, and visualization dashboards (Layer 5).

4.1 RAW SENSOR RECORD SCHEMA (PER NODE TRANSMISSION VIA MQTT)

```
{
  "node_id": "SS-PANEL7-N042",
  "timestamp": "2026-09-09T04:12:33.500Z",
  "gps": { "lat": 23.791204, "lon": 86.433129, "elevation_m": 242.6 },
  "sensors": {
    "tilt_deg": 0.183, // Dual-axis MEMS inclinometer resultant angle
    "vibration_rms_mm_s": 1.42, // Triaxial high-frequency geophone velocity RMS
    "displacement_mm": 3.70, // Extensometer / crack potentiometer movement
    "crack_index": 0.02 // Differential surface shear strain metric
  },
  "node_health": {
    "battery_pct": 78, // LiFePO4 battery state-of-charge
    "rssi_dbm": -71, // LoRa / 802.15.4 link margin
    "hop_count": 3 // Mesh path depth to gateway aggregator
  }
}
```

4.2 FEATURE STORE VECTOR SCHEMA (ENGINEERED WINDOW: 5-MIN ROLLING, 30S STRIDE)

Feature calculation computes descriptive moments, linear slopes, and frequency spectrum energy across a moving window:

FEATURE CATEGORY	COMPUTED VECTOR VARIABLES	PHYSICAL INTERPRETATION IN STRATA CAVING
Kinematic Deformation	tilt_mean, tilt_std, tilt_slope_dt, disp_max, disp_rate_mm_h	Quantifies persistent angular sag and bed displacement velocity above extracted voids.
Seismic / Vibration Energy	vib_rms_max, vib_spectral_energy_10_50hz, crest_factor	Identifies high-frequency micro-cracking and precursor fracturing in rigid roof rock layers.
Spatial Topology Context	nearest_neighbor_dist_m, dist_to_goaf_edge_m, pillar_stress_index	Locates the sensor relative to high-stress abutment pressure zones and worked-out voids.

4.3 MODEL OUTPUT EVENT SCHEMA (PUBLISHED TO FUSION ENGINE KAFKA TOPIC)

```
{
  "node_id": "SS-PANEL7-N042",
  "window_start": "2026-09-09T04:10:00.000Z",
  "window_end": "2026-09-09T04:15:00.000Z",
  "anomaly_score": 0.86,
  "reconstruction_error": 0.0412,
  "contributing_sensors": ["tilt_deg", "displacement_mm"],
  "model_signature": "isoforest_ae_ensemble_v3.2.1",
  "confidence": 0.79,
  "inference_latency_ms": 14.2
}
```

5. COMPONENT TECHNICAL SPECIFICATIONS

5.1 Unsupervised Anomaly Detection (Per-Node Baseline Filter)

CLOUD & EDGE

Objective: Detect statistically anomalous sensor trajectories without requiring historical labels of catastrophic mine subsidence events.

Algorithmic Strategy: Deploys an ensemble of an *Isolation Forest* (100 estimators, sub-sampling ratio 0.75) and a shallow *1D-Convolutional Autoencoder* (latent dimension = 4, LeakyReLU activations, MSE loss). While the Isolation Forest isolates extreme outliers in rolling statistical moments (mean tilt, velocity spikes), the Autoencoder catches subtle, coupled multi-sensor drift where individual channels remain inside statistical ranges but their joint physical correlation breaks down.

The composite per-node anomaly score S_{node} is calculated as: $S_{node} = \alpha \cdot S_{IF} + (1 - \alpha) \cdot \tanh(\beta \cdot MSE_{AE})$ (with default calibrated weights $\alpha = 0.55$, $\beta = 12.5$).

Input Data:	12-dimensional engineered rolling feature vector (Section 4.2) normalized via RobustScaler.
Output Data:	Normalized score S_{node} in $[0.0, 1.0]$, contributing sensor vector, anomaly flag.
Retraining:	Weekly baseline update using moving 30-day non-anomalous baseline window; instant retrain if node is physically relocated.

5.2 LSTM-Based Deformation Progression Forecasting

CLOUD ONLY

Objective: Distinguish high-frequency transient anomalies (e.g., surface compaction from heavy haul trucks or nearby blasting) from continuous, accelerating strata subsidence.

Model Architecture: A two-layer stacked Sequence-to-Sequence (Seq2Seq) Long Short-Term Memory network with 64 hidden units per layer, coupled to a Multi-Head Quantile Regression decoder outputting the 10th, 50th (median), and 90th percentile forecast trajectories across a 24-hour to 72-hour horizon ($\Delta t = 1 \text{ ext{ hour} }$).

Trend Classification: The forecast slope $\frac{d\hat{y}}{dt}$ is evaluated over the horizon: *Class = Accelerating if $d^2y/dt^2 > \theta_{accel}$; Sustained if $dy/dt > \theta_{vel}$; Stable otherwise.*

Input Data:	Past 48 hours of sequential 15-minute aggregated tilt, displacement, and seismic energy readings.
Output Data:	Predicted displacement curves $\hat{y}_{t+h}^{(q)}$ for $q \in [0.1, 0.5, 0.9]$, trajectory classification tag.
Retraining:	Monthly fine-tuning using site historical trends and geotechnical numerical simulations (FLAC3D/UDEC outputs).

5.3 Kriging-Based Spatial Interpolation & Risk Surface

CLOUD PIPELINE

Objective: Transform discrete point observations from mesh nodes into an uninterrupted risk raster across the entire mine panel concession.

Geostatistical Methodology: Deploys *Universal Kriging* with a secondary drift term incorporating mining geometry:

$$Z(s) = m(s) + \epsilon(s), \text{ where } m(s) = \beta_0 + \beta_1 \cdot \text{DistToGoaf}(s) + \beta_2 \cdot \text{OverburdenDepth}(s)$$

The spatial variogram (Spherical or Matérn 3/2 model) is dynamically fitted to empirical covariance every sync interval. Alongside the predicted deformation field $\hat{Z}(s)$, Kriging computes the exact kriging estimation variance $\sigma^2_K(s)$, highlighting sparse zones where risk estimates carry high uncertainty.

Input Data:	GPS coordinates (x_i, y_i) , per-node displacement values, and underground panel boundary vector polygons.
Output Data:	GeoTIFF / GeoJSON risk grid (10m x 10m cells) covering the mine concession + spatial uncertainty raster.
Execution:	Near-real-time recomputation triggered upon arrival of each gateway telemetry batch (≤ 30 seconds).

5.4 Cross-Sensor & Cross-Node Correlation Engine

RULE + STATISTICAL LAYER

Objective: Enforce mandatory physical multi-channel consistency checks before an anomaly is allowed to escalate.

Corroboration Logic Rules:

- Cross-Sensor Coherence (Within Node):** An anomaly requires concordant drift across at least two physically correlated modalities (e.g., $\Delta \text{ext{ tilt} } > 0.15^\circ$ AND $\Delta \text{ext{ displacement} } > 1.5 \text{ ext{ mm} }$). Isolated single-channel spikes are attributed to sensor malfunction or hardware fatigue and redirected to the *Predictive Maintenance Diagnostic Queue*.
- Spatio-Temporal Coincidence (Across Nodes):** Verifies whether neighboring nodes within radius $(R \leq 120 \text{ ext{ m} })$ exhibit statistically correlated anomaly onset within a sliding temporal window $(\Delta t \leq 45 \text{ ext{ minutes} })$. Isolated single-node anomalies are down-weighted by a factor of (0.25) .

Input Data:	Per-node anomaly scores, multi-channel sensor vectors, physical network adjacency matrix.
Output Data:	Corroboration Coefficient $C_{corr} \in [0.0, 1.0]$; Sensor Fault vs. True Ground Movement classifier flag.

5.5 Graph Neural Network (GNN) Mesh Correlation Model

CORE CLOUD DIFFERENTIATOR

Objective: Model non-isotropic subsidence trough propagation across the wireless mesh network, accounting for fault lines, joint orientations, and underground extraction direction.

Architecture: A 3-layer *Graph Attention Network (GATv2)*. The graph $G = (V, E)$ represents sensor nodes as vertices (V) and physical/structural proximity as edges (E) . Edge feature vectors $(\mathbf{e}_{ij})$ encode inter-node distance, elevation difference, geological fault intersection flags, and relative angle to the active longwall face retreat vector.

Attention coefficients (α_{ij}) dynamically learn the spatial transmission paths of ground movement:

$$\alpha_{ij} = \exp(\text{LeakyReLU}(a^T [W \cdot h_i \parallel W \cdot h_j \parallel W_e \cdot e_{ij}])) / \sum_{k \in N(i)} \exp(\text{LeakyReLU}(a^T [W \cdot h_i \parallel W \cdot h_k \parallel W_e \cdot e_{ik}]))$$

Nodes embedded in an actively propagating deformation zone receive reinforcement from neighboring features, separating genuine regional subsidence from local point disturbances.

Input Data:	Node feature matrices $(\mathbf{H} \in \mathbb{R}^{N \times D})$, Graph adjacency edge indices, and edge attribute tensors.
Output Data:	Graph-Refined Risk Score $(R_{\text{GNN}}(v_i) \in [0.0, 1.0])$, Dynamic attention weight tensor matrix $(\mathbf{A})$.
Training:	Pre-trained on geotechnical finite-element subsidence simulations (2,500 synthetic scenarios); fine-tuned on real field data.

5.6 Confidence Scoring & Operational Explainability Layer

AUDIT & DGMS COMPLIANCE

Objective: Eliminate "black-box" resistance from mine safety officers by providing calibrated statistical confidence and plain-language causation narratives.

Confidence Formulation:

$$\text{Confidence} = w_1 \cdot \text{Agreement}(\text{IsoForest}, \text{LSTM}, \text{GNN}) + w_2 \cdot C_{\text{corr}} + w_3 \cdot Q_{\text{mesh}}$$

where (Q_{mesh}) represents packet delivery completeness and battery stability.

Explainability Generation: Computes TreeSHAP feature contributions on the tabular anomaly models and extracts top GAT edge attention weights. A natural language synthesis module compiles an immediate human-readable summary:

"CRITICAL WARNING (Zone 3B, Panel 7): Triggered by sustained tilt surge (+0.24° over 30 min) and differential displacement (+4.8 mm/hr) at Node SS-PANEL7-N042. Corroborated by high spatial attention ($\alpha=0.88$) with adjacent nodes N041 and N044 along the active extraction face. InSAR macro-crosscheck confirms 12mm historical depression basin."

5.7 Predictive Time-to-Critical (TTC) Countdown Model

CLOUD REAL-TIME SERVICE

Objective: Provide an actionable countdown window to structural collapse or critical threshold exceedance for safe evacuation and equipment salvage.

Mathematical Formulation: Let (D_{crit}) be the site-configured critical deformation threshold (e.g., 25mm displacement or 1.5° angular tilt). The model projects the LSTM quantile trajectories $(\hat{y}^{(q)}(t))$:

$$TTC_{\text{critical}} = \inf \{ t > t_0 : \hat{y}(t) \geq D_{\text{crit}} \} \longrightarrow [TTC_{\text{min}}, TTC_{\text{median}}, TTC_{\text{max}}]$$

If the median estimate falls below 8 hours, or if the second derivative $(d^2\hat{y}/dt^2)$ spikes by >40% in two consecutive windows, the system automatically triggers an **Immediate Emergency Escalation**.

5.8 Automatic False-Alarm Feedback & Active Retraining Loop

AUTOMATED PIPELINE

Objective: Autonomously improve detection precision over the lifecycle of the mine without manual data science code modifications.

Workflow: When an alert triggers, dashboard operators can mark: [Confirmed Ground Movement], [False Alarm – Surface Blast], [False Alarm – Machinery Vibration], or [Sensor Hardware Fault]. Confirmed false alarms are automatically tagged and dumped into the negative-mining sample repository.

A weekly automated Airflow DAG retrains local IsoForest contamination rates and fine-tunes GAT attention bias matrices. A strict **Validation Gate** enforces that candidate models must demonstrate equal or superior recall on known subsidence events and ≥35% reduction in verified false positive triggers before being swapped into the active inference endpoint.

5.9 TinyML Edge-Deployment Architecture (Zero-Connectivity Mode)

ESP32 / GATEWAY MICROCONTROLLER

Objective: Maintain autonomous safety alerting when backhaul cellular or satellite links fail during severe weather or underground power interruptions.

Quantization & Execution Profile: The cloud Autoencoder is distilled into a 3-layer MLP, quantized to int8 precision via TensorFlow Lite Micro. The quantized binary occupies <38 KB SRAM and executes in 4.8 ms on an ESP32-S3 (Xtensa dual-core @ 240 MHz). Gateways execute local neighbor corroboration across incoming mesh radio packets. If confirmed, edge relays trigger hardwired local strobes and high-decibel acoustic horns immediately, while appending the incident to a circular SPIFFS flash buffer for synchronization upon backhaul restoration.

5.10 Satellite InSAR Fusion (Macro-Scale Cross-Check)

MACRO VALIDATION LAYER

Objective: Provide an external geodetic validation layer using European Space Agency (ESA) Sentinel-1 Synthetic Aperture Radar interferometry.

Integration Pipeline: Ingests Level-1 Single Look Complex (SLC) interferograms every 6 to 12 days. Line-of-Sight (LOS) velocity displacement rasters are georeferenced and overlaid against the Kriging ground sensor risk surface. A spatial divergence index $\Delta_{\text{InSAR}} = |Z_{\text{sensor}} - Z_{\text{InSAR}}|$ highlights regional subsidence bowls occurring outside the active mesh node footprint, prompting relocation of wireless sensor nodes to cover unmonitored developing troughs.

6. FUSION & ALERT DECISION LOGIC

The Fusion Decision Engine combines outputs from all ten algorithmic sub-modules into three actionable operational tiers. Every tier requires multi-evidence corroboration:

ALERT TIER	TRIGGER CONDITION & EVIDENCE MATRIX	OPERATIONAL ACTION	PROTOCOL
ADVISORY	- Single node anomaly score $S_{\text{node}} > 0.65$ - Single-sensor drift or uncorroborated neighbor status - LSTM trend shows flat or decelerating slope - Confidence score < 0.60	Dashboard notification; automated health ping to check node battery and RSSI.	Level 1 Logging (Audit Trail)
WARNING	$S_{\text{node}} \geq 0.75$ across ≥ 2 channels (tilt + disp) - Neighbor corroboration $C_{\text{corr}} \geq 0.70$ (within 120m) - GNN Risk Score $R_{\text{GNN}} \geq 0.70$ - LSTM classifies movement as <i>Sustained</i>	SMS / Telegram alert to Mine Geotechnical Manager; automated inspection dispatch.	Level 2 Escalation (DGMS Tracked)
CRITICAL	- Warning conditions fulfilled simultaneously - Predictive Time-to-Critical $TTC_{\text{median}} \leq 12 \text{ ext{hours}}$ - LSTM predicts rapid acceleration $d^2y/dt^2 > \text{heta}_{\text{accel}}$ - Or instantaneous discontinuity $\Delta \text{ext{disp}} > 10 \text{ ext{mm}}$	Automated activation of underground audio-visual sirens, conveyor cutoff, zone evacuation.	Level 3 Immediate Life-Safety Protocol

7. TECHNOLOGY STACK SUMMARY

Platform Tier	Selected Technologies	Architectural Rationale
Streaming & Ingestion	Apache Kafka, Eclipse EMQX MQTT Broker	High-throughput pub/sub handling millions of sensor packets/sec with persistent back-buffering.
Time-Series Storage	TimescaleDB / InfluxDB Enterprise	Optimized relational + hypertable storage for rolling aggregations, continuous downsampling, and spatial metadata.
ML Core & Deep Learning	PyTorch 2.3, PyTorch Geometric (PyG), scikit-learn	Native tensor computation for GAT message-passing, LSTM forecasting, and vectorized Isolation Forest pipelines.
Spatial Geostatistics	PyKrig, GSTools, GDAL, GeoPandas	Rigorous 2D/3D ordinary and universal kriging interpolation with custom variogram model fitting.
Explainability & Audit	SHAP (TreeSHAP / DeepSHAP), Captum	Direct model attribution calculation guaranteeing transparent feature contribution outputs.
Edge & TinyML Runtime	TensorFlow Lite Micro, Edge Impulse C++ SDK	Sub-50KB footprint compiled for Xtensa LX7 and ARM Cortex-M4 architectures with SIMD optimization.
Serving & Orchestration	FastAPI, TorchServe, Triton, Apache Airflow	Sub-20ms inference latency, dynamic model batching, and automated validation-gated continuous retraining DAGs.

8. PRODUCTION VERIFICATION & EVALUATION METRICS

Evaluation Metric	Target Threshold	Verification Protocol & Geotechnical Rationale
Precision / Recall / F1	Recall ≥ 0.98, Precision ≥ 0.90	Tested against historical mining collapse databases and validated geotechnical FEA benchmark runs.
Actionable Warning Lead Time	≥ 8.0 Hours Ahead of Collapse	Time delta between first verified <i>Warning</i> alert and physical surface cracking/caving event.
False Alarm Rate	< 0.05 per Node / Month	Ensures mine operators maintain trust in the automated alert system and prevents shutdown fatigue.
Telemetry Packet Delivery	> 96.5% Ingestion Reliability	Mesh QoS tracking ensuring input vector completeness across rolling windows.
Forecast Calibration Error	Expected Calibration Error < 0.08	Validates that stated 90% quantile confidence bands contain ground-truth outcomes at least 90% of the time.
Edge vs. Cloud Parity	> 94.0% Agreement Rate	Measures consistency between localized TinyML edge classifications and full cloud ensemble decisions.

9. OPEN ENGINEERING CONSIDERATIONS & RISK MITIGATION

Key Implementation Risks & Mitigation Strategies

- Cold-Start Initialization on Virgin Panels:** Initial installations lack site-specific empirical deformation baselines. *Mitigation:* Bootstrap using synthetic deformation profiles generated by numerical strata simulation tools (FLAC3D/UDEC) calibrated to regional geological boreholes.
- Geotechnical Heterogeneity Across Coalfields:** Strata stiffness varies drastically between Raniganj sandstone and Jharia shale. *Mitigation:* Enforce panel-cluster model partitioning rather than a universal national weights file.
- Regulatory Audit Compliance (DGMS Mandate):** Automated threshold retraining must not silently alter safety trip points. *Mitigation:* Maintain an immutable, cryptographically signed ledger of all threshold revisions with manual sign-off required for changes >15%.
- Kriging vs. GNN Dynamic Weighting:** Kriging provides smooth mathematical interpolation, while GNN models directional fracture channeling. *Mitigation:* Employ an adaptive entropy-weighted blending algorithm that increases GNN weighting in regions exhibiting strong spatial covariance gradients.

